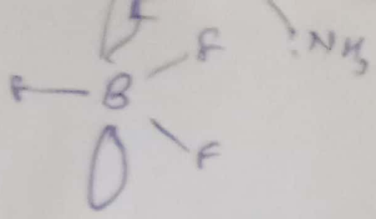

Boron family — p-ගොනුවට අයත්

3 කාණ්ඩය — පොදු ඉ'න වින්‍යාසය $ns^2 np^1$

B A

ද්‍රවාංක	B	Boiling point වැඩිවේ.
අඩුය	Al	
වැඩිවේ	Ga	
Density	In	
වැඩිවේ	Tl	

B.P හා M.P depend වෙන්නේ අන්තර් අණුක ආකර්ෂණ බල මත

* කාණ්ඩයේ
* Electronegativity ඉතා වැඩි ම B
* Densities increase down the group
* B has a

p block elements.

Groups →	13	14	15	16	17	18
						He
	B	C	N	O	F	Ne
	Al	Si	P	S	Cl	Ar
	Ga	Ge	As	Se	Br	Kr
	In	Sn	Sb	Te	I	Xe
	Tl	Pb	Bi	Po	At	Rn

Non Metals

Metals

Metalloids

- Group 17 Halogens
- Group 18 Noble Gases

The Boron Family

ආවර්තිතා වගුවේ 3A (13) වන කාණ්ඩය

පොදු e ක වින්‍යාසය $ns^2 np^1$

+3 oxidation state ඇති කාණ්ඩයකි

තම මූලද්‍රව්‍යයන්ට තවත් ඔක්සිකරණ වේ. කාණ්ඩයේ

පහළට යන විට ද +1 ක ඔක්සිකරණ පෙන්වයි.

වඩා

2A (2) elements වලට Electro negativity විද්‍යුත් සෘණතාවය වැඩියි.

2A (2) elements වලට වඩා atomic size පරමාණුවේ ප්‍රමාණය කුඩායි.

because it is easier to remove an electron from the higher energy p sublevel.

මෙහි ගොනු e.c වින්‍යාසය $ns^2 np^1$

$\boxed{\uparrow\downarrow}$ $\boxed{\uparrow}\boxed{}\boxed{}$

remove කරන්නේ මෙයයි

p sublevel

* Atomic size, Ionization Energy, and Electro Negativity do not change as expected down the group

කාණ්ඩයේ පහළට අපේක්ෂිත රටාව අනුව AS, IE, EN අගය වෙනස් වෙන්නේ නෑ.

Because there are intervening transition and inner transition elements

(ආවර්තයේ මැද d. ගොනුව පිහිටීම කාණ්ඩ 2ක් අතර පරතරය ඉලෙක්ට්‍රෝන පුරවන ලෙස ඇති නිසා)

* B has a much higher melting point and boiling point than others.
* Boiling points decrease down the group
But there is no overall trend for melting point.

* Densities increase down the group

~~Hence~~ * Boiling point should decrease down the group from B to Tl.
The forces of attraction decrease down the group because of the increase in size of atoms down the group. Hence, the boiling point decreases down the group.

Melting point → ?

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* Why density increase down Group 13?

Density is mass per volume.

The effect of increased atomic mass is more as compared to increased size. Therefore, the effect of increase in mass is more than increase in volume.

* The Redox behavior of the elements in this group

1. multiple oxidation states present.

2. +1 , +3

2. Lower state becomes more prominent going down the group.

+1 ඔක්සිකරණ අංකය ස්ථායීවන්නේ Tl වල

3. +1 ට අදාල අයන oxides, +3 ට අදාල අයන oxides වලට වඩා basic ගුණයෙන් යුතුය.

Oxides with the element in a lower oxidation state are more basic than oxides with the element in a higher oxidation state

$Ga_2O > Ga_2O_3$

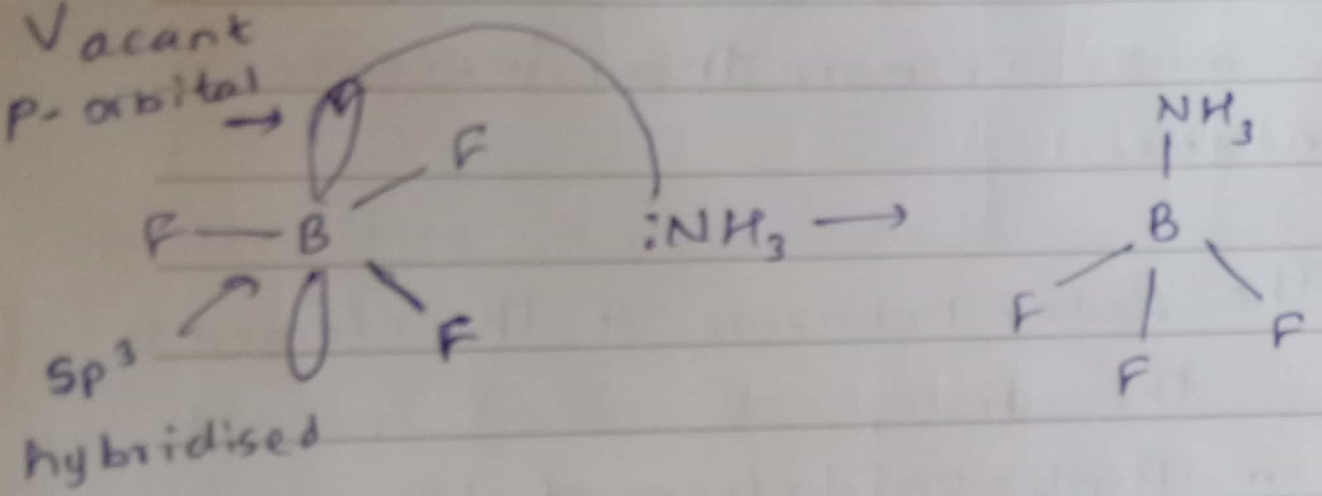
$\xleftarrow{\hspace{2cm}}$

more basic

* Accepting a Bonding Pair from an Electron-Rich Atom

Gaseous $BX_3 \longrightarrow BF_3$

B atom is electron deficient [illegible]



$BF_{3(g)} + :NH_{3(g)} \longrightarrow F_3B—NH_{3(g)}$

මෙවැනි reactions වලට කියන්නේ Lewis acid-base reactions කියා.

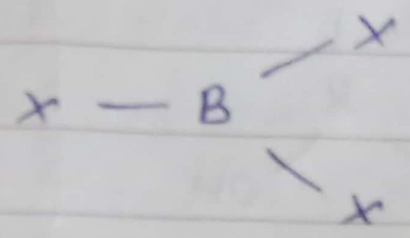
සහසංයුජ බන්ධනයක් සෑදීම සඳහා එක් ප්‍රතික්‍රියකයක් e' ක යුගල අනෙකෙන් ලබා ගන්නා.

Lewis acid-base reactions are in which one reactant accepts an electron pair from another to form a covalent bond

B is a Lewis acid

B has no tendency to form ions, and always forms covalent bonds

BF_3 - Gaseous
BCl_3 - Liquid
BI_3 - Solid



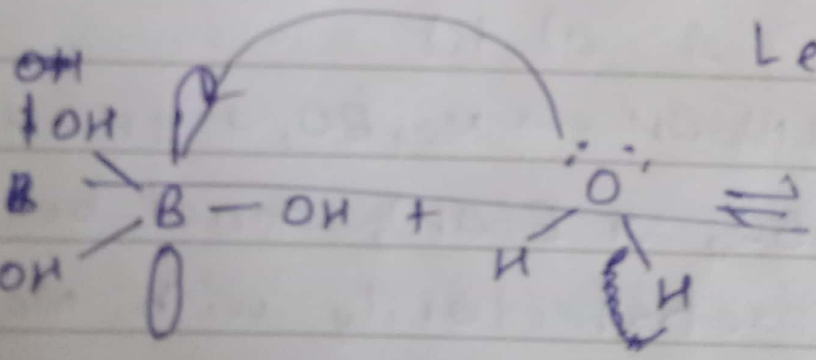
X = F, Cl, Br

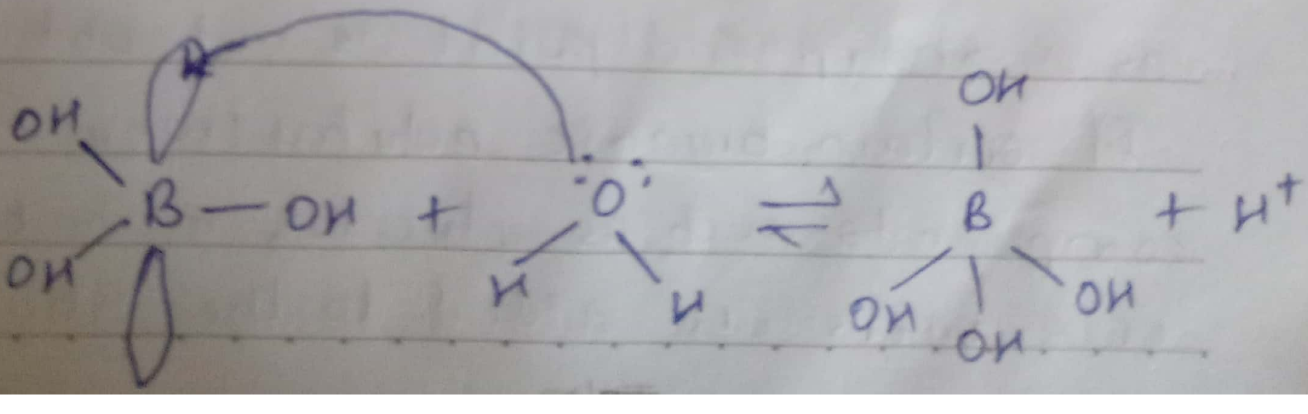
planar triangle 120°

B_2O_3 - acidic oxide

$B(OH)_3$ boric acid in water, the acid itself does not release a proton.

Lewis acid





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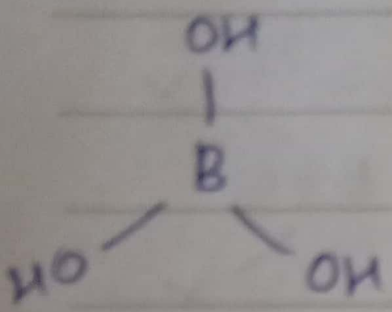
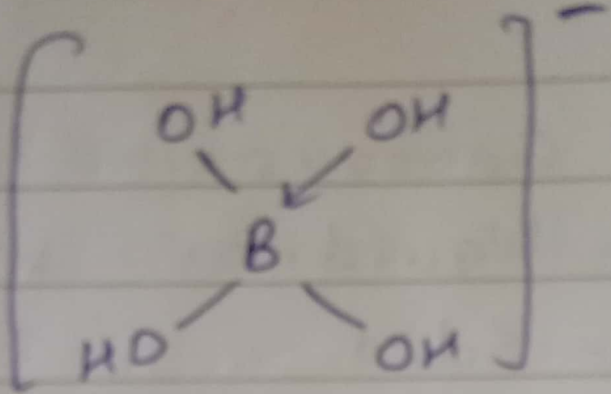
$B(OH)_{3(s)} + H_2O_{(l)} \rightleftharpoons B(OH)^-_{4(aq)} + H^+_{(aq)}$

$[H_3BO_3]$





Plane triangle

tetrahedral metaborate ion

$$H_3BO_3 \rightleftharpoons H^+ + H_2BO_3^-$$

$$H_3BO_3 + 3NaOH \longrightarrow Na_3BO_3 + 3HOH$$

2019 Q A a) ii)

$H_3BO_3 + 3NaOH \longrightarrow Na_3BO_3 + 3HOH$

Thus H_3BO_3 or $B(OH)_3$ cannot be titrated satisfactorily with NaOH, as a sharp end point is not obtained. If certain organic polyhydroxy compounds such as glycerol, mannitol or surgers are added to the titration

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mixture, then $B(OH)_3$ behaves as a strong monobasic acid

It can now titrated with NaOH, and the end point is detected using phenolphthalein as indicator

$$B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4]$$

$$NaBO_2 + 2H_2O$$

Sodium metaborate

H_3C-CH_3 (ethane) and H_3B-NH_3 (amine-borane) are isoelectronic

C has 4 valence electrons $\cdot\dot{C}-\dot{C}\cdot$

and $\cdot\dot{B}-\dot{N}\cdot$ N has five

have the same number of valence e'ns

Boron nitride very similar to Graphite like graphite, its e'ns are highly delocalized through resonance

[illegible]

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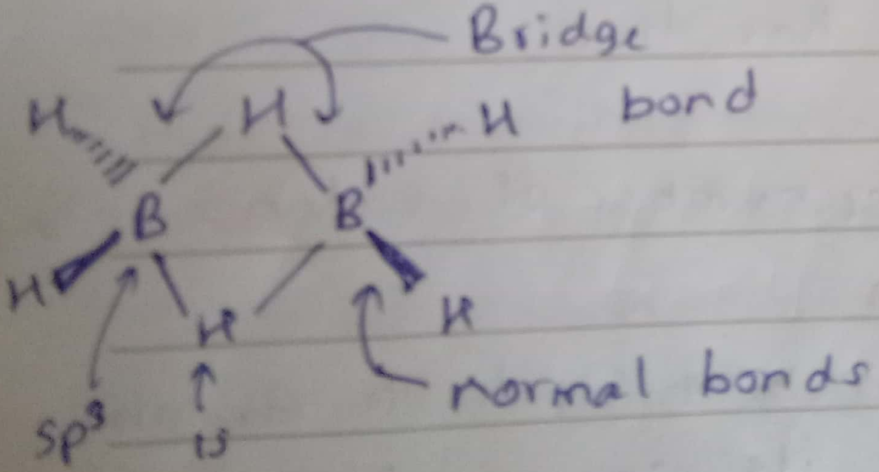
Forming Bridge Bonds with Electron-Poor Atoms

In elemental boron සමේම්නා B සහ ශිනි ගොනුවක් hydrides (boranes) වලට electron-rich atom නෑ e^-ns supply කරන්න. ∴ B attains an octet ලබාගැනීමට සිදුවීම නිසා හරිත unusual bonding

diborane (B_2H_6) and many larger boranes

2 types of B-H bonds exist

1. A typical electron-pair bond.



2021 A (a)

2018 A (D)

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2. Hydride Bridge bond
(three-center, two electron bond)

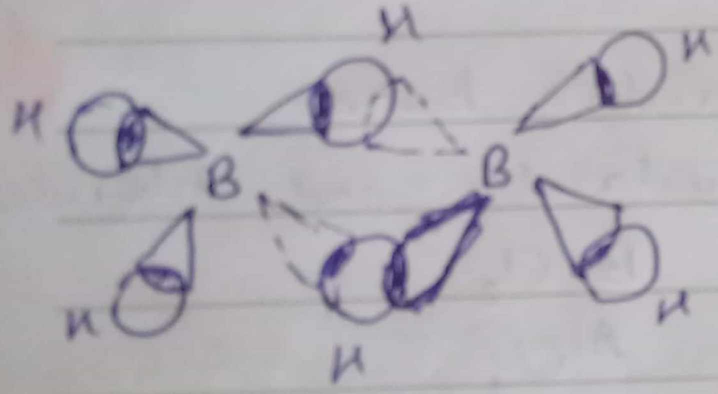
Each B–H–B grouping is held together by only two electrons

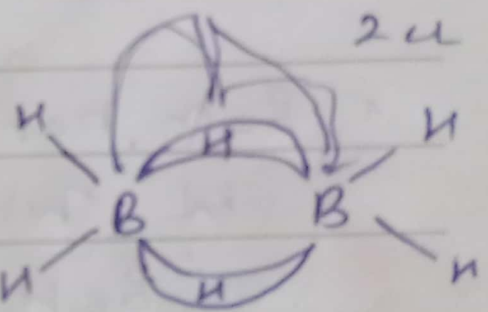
Eg. B_{12} unit

B_5H_9

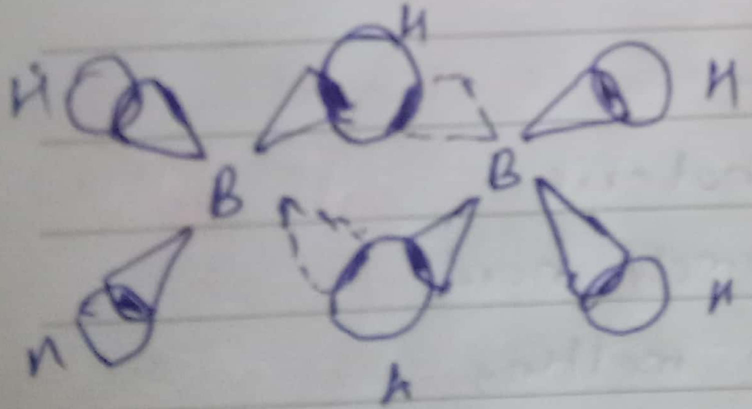
3 centres

2 e





banana bonds



Diagonal Relationships → Be, Al

~~Be~~ B
~~Al~~ Si

Be B C
Al Si

* Oxoanions in strong base
  → $Be(OH)_4^{2-}$ beryllate
  → $Al(OH)_4^{2-}$ Aluminate

* Both have bridge bonds in their hydrides and chlorides

  BeH_2 $BeCl_2$
  AlH_3 $AlCl_3$

* Oxides both
  amphoteric
  extremely hard
  high melting

* Differ → atomic and ionic sizes

small highly charged

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Important Reactions

① react sluggishly with water

$2Ga_{(s)} + 6H_2O(hot) \rightarrow 2Ga^{3+}_{(aq)} + 6\bar{O}H_{(aq)} + 3H_{2(g)}$

$2Tl_{(s)} + 2H_2O(steam) \rightarrow 2Tl^{+}_{(aq)} + 2\bar{O}H_{(aq)} + H_{2(g)}$

② When strongly heated in pure O_2

All members of oxi (E = B, Al, Ga, In)

$4E_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2E_2O_{3(s)}$

$4Tl_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2Tl_2O_{(s)}$

$B_2O_3 > Al_2O_3 > Ga_2O_3 > In_2O_3 > Tl_2O$

weakly acidic → strongly basic

Oxide acidity decreases down the group

+1 oxide > +3 oxide

← more basic

3) Am are reduce halogens (X_2)

$2E_{(s)} + 3X_2 \longrightarrow 2EX_3$ $(E = B, Al, Ga, In)$

$2Tl_{(s)} + X_2 \longrightarrow 2TlX_{(s)}$

$BX_3 \rightarrow$ volatile

covalent

4) Acid treatement of Al_2O_3 c water

purification

$Al_2O_{3(s)} + 3H_2SO_{4(l)} \longrightarrow Al_2(SO_4)_{3(s)} + 3H_2O_{(l)}$

5) Redox

$2Al_2O_{3(s)} + 3C_{(s)} \longrightarrow 4Al_{(s)} + 3CO_{2(g)}$

Electrochemically → with cryolite

with lower MP

(Na_3AlF_6)

$2(\;Al_2O_3 + 6H^+ + 6e \longrightarrow 2Al + 3H_2O$

$3(2H_2O + C \rightarrow CO_2 + 4H^+ + 4e$

$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$

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6) Displacement Gallium Arsenide
GaAs

$(CH_3)_3Ga_{(g)} + AsH_{3(g)} \rightarrow 3CH_{4(g)} + GaAs_{(s)}$

Important compound

1) Boron oxide B_2O_3 → Production of borosilicate glass

2) Borax $Na_2[B_4O_5(OH)_4].8H_2O$ fireproof insulation material and as a washing powder

3) Boric acid H_3BO_3 / $B(OH)_3$

4) Diborane B_2H_6

5) Aluminium Sulfate (cake alum) $Al_2(SO_4)_3$

6) Al_2O_3

7) $Tl_2Ba_2Ca_2Cu_3O_{10}$

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Carbon Family 4A (14)

~~ns^2np~~ $ns^2 np^2$

H

He

Li Be B C

$\boxed{\uparrow\downarrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{\ }$

	M.P (°C)	
C	4100	large decrease, due to longer, weaker bonds in Si
Covalent network → Si	1420	
Ge	945	due to change from Si struct covalent network to metallic bond
metallic bonding → Sn	232	
Pb	327	

	MP		MP
Al	660	Si	1420
Metallic	→	~~Metalloid~~ Covalent network	
Ga	30	Ge	945

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Allotropism - බහුරූපී ආකාර
different crystalline or molecular forms of a substance.

[illegible] උෂ්ණ දී හා පීඩනයේ දී එකක් අනෙකට වඩා stable

C

Graphite	more stable > Diamond
black	colorless
electrical conductor	electrical insulator

Sn

white-tin	Gray-tin
stable (room T)	Stable in below 13°C

$2s^2 2p^2$	C	* number of
	-4, +4, +2	oxidation states
$3s^2 3p^2$	Si	decreases.
	-4, +4	the lower (+2) state
$4s^2 4p^2$	Ge	becomes more common
	+4, +2	
$5s^2 5p^2$	Sn	* Atomic size increases
	+4, +2	
$6s^2 6p^2$	Pb	* B.P decrease
	+4, +2	* Density increases
		(crystal packing)

IE and EN do not decrease smoothly because transition and inner transition elements intervene

Solid [illegible] type [illegible] down the group hardness and melting point decrease

C
Si
Ge } Bonding type - Network covalent

Sn
Pb } metallic

Important Reactions

1) Oxidized by halogens:

$E_{(s)} + 2X_2 \longrightarrow EX_4$ ($E = C, Si, Ge$)

+2 halides more stable for Sn, Pb

2) Oxidized by O_2

$E_{(s)} + O_{2(g)} \longrightarrow EO_2$

Pb → +2 → PbO

Oxides become more basic down the group

Provide weak acidity of natural unpoluted water

$CO_{2(g)} + H_2O_{(l)} \rightleftharpoons [H_2CO_{3(aq)}]$

$\rightleftharpoons H^+_{(aq)} + HCO_3^-{}_{(aq)}$

hot coke

3) $C_{(s)} + air_{(g)} + H_2O_{(g)} \longrightarrow CO_{(g)} + CO_{2(g)} + N_{2(g)} + H_{2(g)}$

4) Hydrocarbons → heat or electricity

$CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(g)}$

5) $CaC_{2(s)} + 2H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)} + C_2H_{2(g)}$

Carbide   Acetylene

$C_2H_{2(g)} \rightarrow$ used to make other organic compounds

fuel in welding

6) $CCl_{4(l)} + HF_{(g)} \longrightarrow CFCl_{3(g)} + HCl_{(g)}$

Freons (Freon-11)

Chlorofluorocarbons

7) $SiO_{2(s)} + 2C_{(s)} \longrightarrow Si_{(s)} + 2CO_{(g)}$

Silica   Silicon

Computer chips

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Important Compounds

1) CO
2) CO_2
3) CH_4
4) SiO_2
5) SiC
6) R_4Sn Organotin compound
7) Tetraethyl lead, $(C_2H_5)_4Pb$

Highlights of Carbon Chemistry

* C forms bonds with
  1A (1), 2A (2) metals
  many transition metals
  Halogens
  B, Si, O, N, P, S

* Every oxidation state posible
  $+4 \rightarrow -4$
  CO_2 CH_4

* Ability to bond to itself
  To form chains
  To form multiple bonds

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$2s^2 \ 2p^2$ $\boxed{\uparrow\downarrow}$ $\underbrace{\boxed{\uparrow}\boxed{\uparrow}}_{\pi \text{ bonds form}}\boxed{\ }$

C – C

Si — Si

Ge — Ge

E – E bonds become longer and weaker → bond strength ↓

* Carbonates are used in several common antacides

$CaCO_{3(s)} + \underset{\text{in stomach acid}}{2HCl_{(aq)}} \longrightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$

✓ 4A → solid network-covalent or ionic oxide

CO_2 , CO gaseous oxides

* CN^- is isoelectronic (same electronic structure) as CO

$[:C \equiv N:]^-$ $:C \equiv O:$

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මෙන්ම සංයුජ ශක්ති හිස් d කාක්ෂික
Si වල ඇති නිසා

Si halide > C halide
$\longleftarrow$
more reactive

$p_\pi - d_\pi$ bond → overlap Si d
halogen p

Silicate minerals

Oxygen :- the most abundant element on Earth
Silicon - Next abundant

Silicone polymers

$$Si_{(s)} + 2CH_3Cl_{(g)} \xrightarrow[\sim 300^\circ C]{Cu\ catalyst} (CH_3)_2SiCl_{2(g)}$$

$$(CH_3)_2SiCl_{2(l)} + 2H_2O_{(l)} \longrightarrow (CH_3)_2Si(OH)_{2(l)} + 2HCl_{(g)}$$

$$n(CH_3)_2Si(OH)_2 \longrightarrow \left[-O-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{Si}}}}- \right]_n + nH_2O$$

poly(dimethyl siloxane)

Diagonal Relations ... B and Si ...

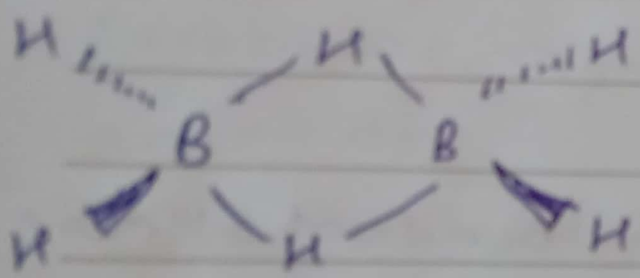
B C

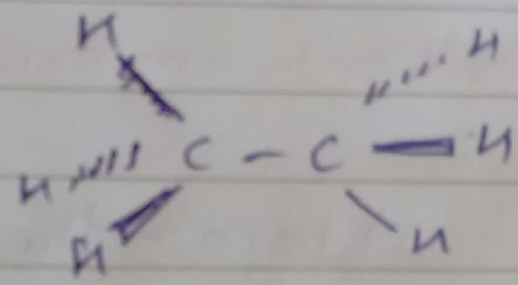
Al Si

Both.

The electrical properties of a semiconductor

2011 4 (a)





B_2H_6 C_2H_6

diborane ethane

2 e'ns යි

B හි valence e'n's 3 යි.

C හි val 4 යි.

2021

4. b) Be Mg දෙකම 2(A) කාණ්ඩයේ

Be ඉලෙක්ට්‍රෝන වින්‍යාසය $1s^2\ 2s^2$

Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$

Be හි d subshells උප කාක්ෂික නැ.

$\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ 4 valency e ය.

coordination number 4

Mg d subshells පවතී.

$\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}$ $\boxed{3s^1}$

$2s^2$ $2p^6$

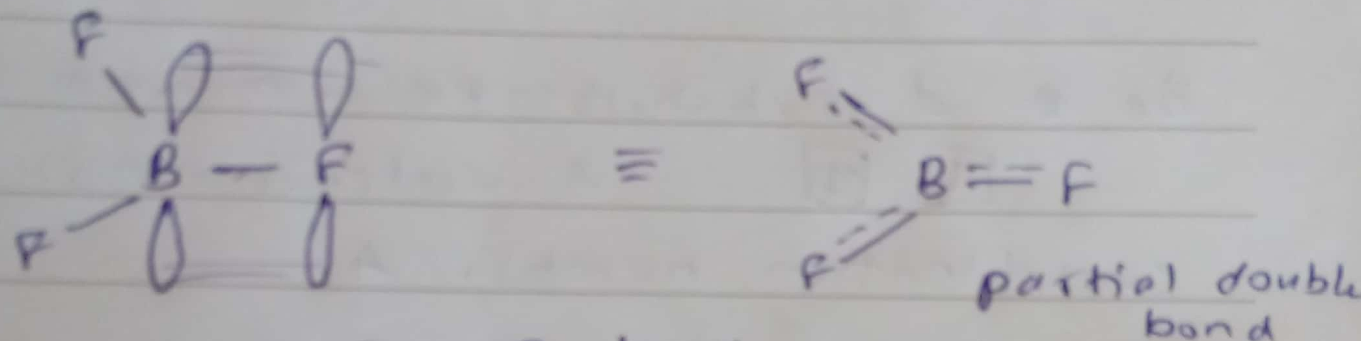
3 වන ශ.මට්ටමේ d හිස් කාක්ෂික තිබේ,

∴ Mg shows hexahydrated complexes

2018 . . . .

4.(b) expected B-F bond length එකට වඩා අඩු නිරීක්ෂිත bond length එකක් ලැබේ.

$p\pi - p\pi$ back bonding



$p\pi - p\pi$ back bonding

B හි electron deficiency එක එය ඉවත් කරගනී

4. (c) Noble gas have electronic configuration $1s^2\ 2s^2\ 3p^6$

Monoatomic

no interatomic forces

∴ weak dispersion forces පවතියි

∴ very low tempreture ~~යටතේ~~ liquified කළ හැක.

H							
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca						

He ~~He~~

2019

1. a) (ii) H_3BO_3 is a very weak acid and direct titration with NaOH is not possible. However in the presence of organic polyhydroxy compounds such as glycerol, mannitol ets, facilitates the acid-base titration. (20)

b) Discuss the properties of trimethyl amine $(CH_3)_3N$ and trisilylamine $(SiH_3)_3N$ with respect to their molecular structures. (20)

c) $SiCl_4$ rapidly hydrolyses in aqeous alkaline media whereas SiF_4 doesn't. Explain with appropriate reactions. (20)

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d) What would you observe if an aqueous solution of Na_2CO_3 is added to an aqueous solution of $AlCl_3$. Explain your answer

2021

4. a) Describe the structural difference between B_2H_6 and C_2H_6. Explain with illustrations. (30)

b) Be and Mg are both group 2 elements. Among the two elements Mg shows hexahydrated complexes, whereas Be does not. Justify your answer (20)

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2018

A. C) Why do the noblet gases have low boiling points? (16)

D) Explain the nature of bonding of the molecule B_2H_6. Include the orbital picture in your answer (20)

E) Bond length of the BF_3 molecule is shorter than the expected B-F bond length. Clarify this statement using your knowledge of backbonding (20)

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